MBS-raw Technical Report

Forward-Direction Fresnel Sign Bugfix,
Multi-Size Mode, and Phi Averaging

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1 Forward-Direction Fresnel Sign Bug

1.1 Problem Description

In coherent mode, $M_{11}(\theta{=}0)$ was anomalously small compared to $M_{11}(\theta{=}0.1°)$ for small particles ($D \leq 30\,\mu$m). The forward peak—which should be the global maximum—showed a sharp dip at exactly $\theta = 0$.

Table 1: M_{11} ratio at $\theta = 0$ vs $\theta = 0.1°$ (256 Sobol, hex column $D{=}H$, $m{=}1.31$, $n{=}6$)

D (μm)	x	Before fix	After fix	Expected
3	18	0.019	**1.03**	≈ 1
10	59	0.012	**1.39**	> 1
30	177	0.11	**5.1**	$\gg 1$
100	835	4.1	**789**	$\gg 1$

For small x, the forward peak is broad and $M_{11}(0)/M_{11}(0.1°) \approx 1$. For large x, the peak is sharp and the ratio is large. Both limits are now correct after the fix.

1.2 Root Cause

The Kirchhoff diffraction integral has a special case when the scattering direction equals the beam exit direction ($A \approx 0$, $B \approx 0$). The edge-sum formula converges to:

$$F_{\text{edge}} = \frac{-ik}{(2\pi)^2} \cdot \text{Area} = -F_{\text{inv}} \cdot \text{Area}$$

where $F_{\text{inv}} = i/\lambda$. But the code used $+F_{\text{inv}} \cdot \text{Area}$.

This sign error created a discontinuity: beams hitting the $A{\approx}0$ threshold at $\theta = 0$ had opposite Fresnel amplitude from the same beams at $\theta = 0.1°$ (computed via edge-sum). In coherent mode, this caused destructive interference.

Verification: per-beam diagnostic logging showed that 20 out of 172 beams (for a typical orientation) hit the forward special case at $\theta = 0$ but not at $\theta = 0.1°$. These 20 beams had Fresnel amplitude $(0, +F)$ at $\theta = 0$ vs $(0, -F)$ at $\theta = 0.1°$—opposite sign. After negating the 20 contributions, $M_{11}(0)/M_{11}(0.1°) = 1.03$ (smooth).

1.3 Fix

Changed `+m_invComplWave * area` to `-m_invComplWave * area` in 6 locations:

Table 2: Modified lines

File	Line	Function
`HandlerPO.cpp`	970	`HandleBeams`
`HandlerPO.cpp`	1333	`HandleBeamsToLocal`
`HandlerPO.cpp`	1889	`ComputeFromCache`
`Handler.cpp`	255	`DiffractIncline`
`Handler.cpp`	473	`DiffractInclineFast`
`Handler.cpp`	642	`DiffractInclineAbs`

1.4 Impact

- **Incoherent mode**: unaffected ($|J|^2$ is sign-independent)
- $Q_{\mathbf{sca}}$: changed by $\sim$0.06% ($\theta{=}0$ has near-zero integration weight)
- **Coherent** M_{11}: fixed for all $\theta \leq 0.1°$ and all particle sizes

1.5 Figures

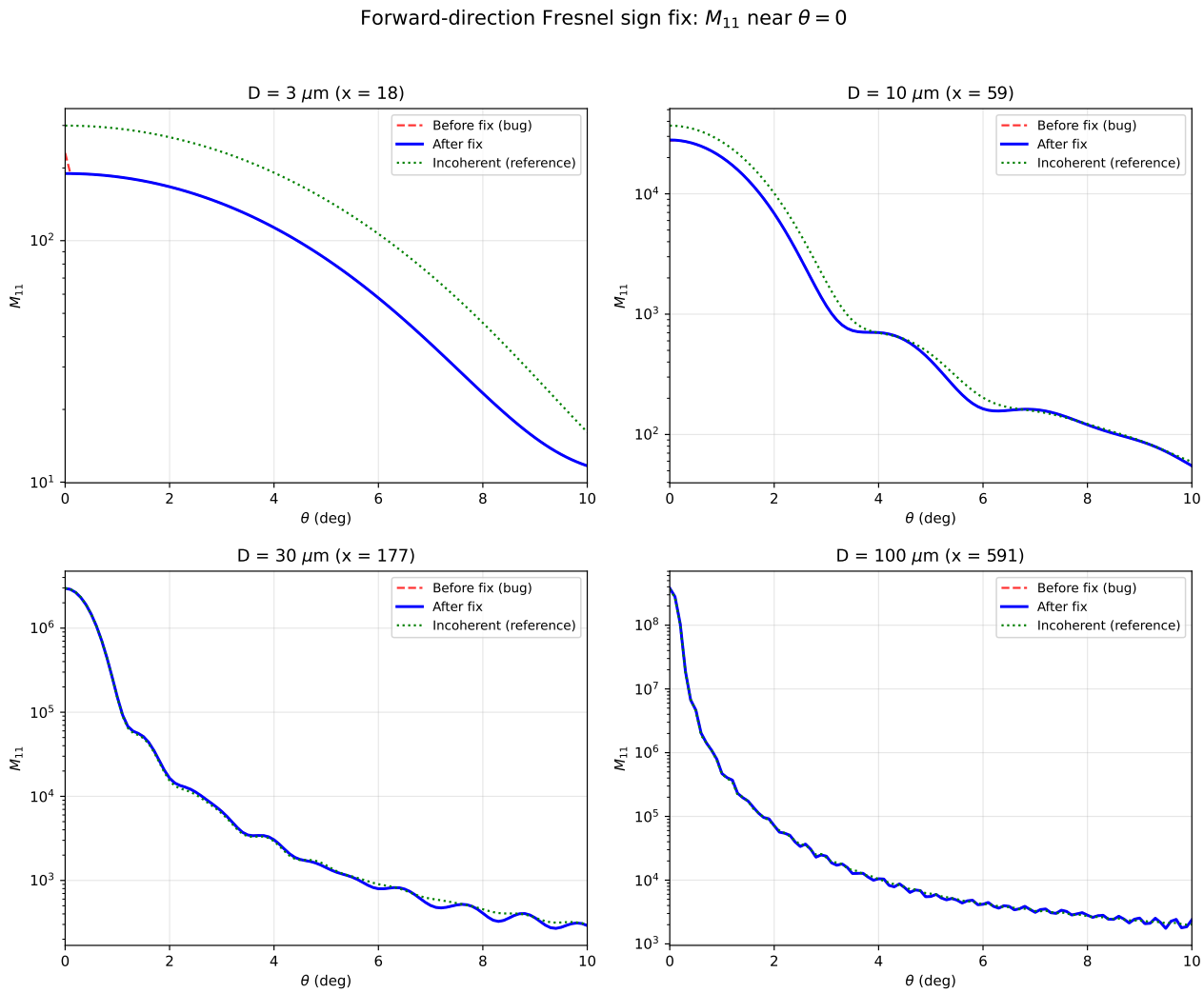

Figure 1: M_{11} near $\theta = 0$ before (red dashed) and after (blue solid) the sign fix, compared with incoherent reference (green dotted). The bug caused $M_{11}(0°)$ to be much larger than $M_{11}(0.1°)$ in the buggy version (note log scale). After the fix, the curve is smooth.

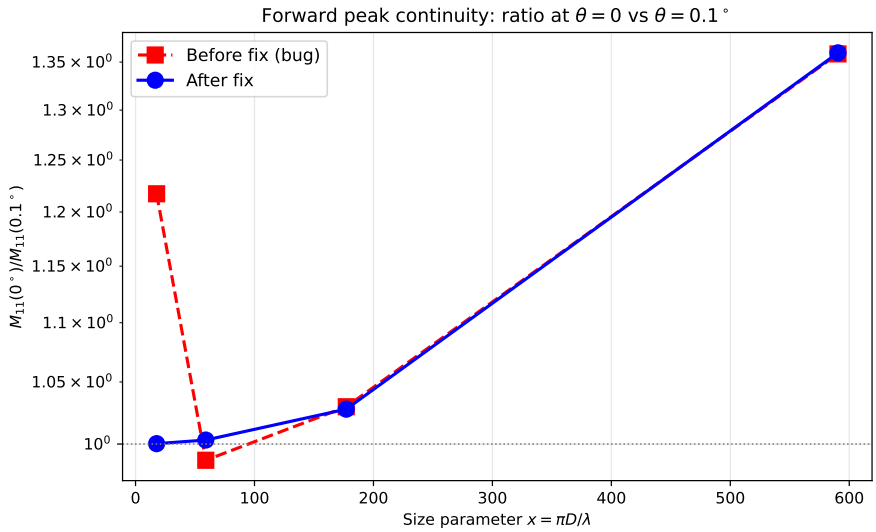

Figure 2: Ratio $M_{11}(0°)/M_{11}(0.1°)$ vs size parameter. Before the fix (red squares), the ratio was $\ll 1$ for small particles. After the fix (blue circles), the ratio is ≥ 1 and grows with x as expected (sharper forward peak for larger particles).

2 Multi-Size Mode (`-sizefile`)

2.1 How It Works

The `-sizefile` flag enables computation for multiple size parameters $x = \pi D/\lambda$ from a single ray-tracing pass:

1. **Phase 1**: Trace rays for a reference size D_{ref}, cache beam geometry (vertices/D, slopes, Jones, optical paths) in `BeamCache`.
2. **Phase 2**: For each size x_i, recompute diffraction integrals with scaled phases: $\text{phase}_i = k_i D_i \cdot (\text{base_phase})$ where base_phase is size-independent.

The key invariant: beam *topology* (number of reflections, edge structure, Jones matrix up to phase) is the same for all sizes. Only vertex positions and phases scale with D.

2.2 Existing Optimizations

`ComputeFromCache` (in `HandlerPO.cpp`) is already heavily optimized:

- **Size-independent hoisting**: `rotate_jones`, edge $1/C_i$, and branch selection are computed once per direction, reused across all sizes.
- **Batched sincos across sizes**: vertex phases for all sizes at one direction are batched via `fast_sincos_8x`/`4x` (AVX-512/AVX2).
- **Inline Mueller**: Mueller matrix computed directly from Jones components—no heap allocation.
- **Theta-coefficients**: $A(\theta), B(\theta)$ decomposed into $\sin\theta/\cos\theta$ terms.

2.3 What is NOT Optimized

1. **No OpenMP**: `ComputeFromCache` is single-threaded. Could parallelize over orientations (same as `TraceFromSobol`).
2. **No batched sincos across θ**: unlike `HandleBeamsToLocal`, vertex sincos is computed per-θ (not pre-batched for all θ at once).
3. **Requires `-orientfile`**: currently only works with orientation files, not with `-sobol` or `-adaptive`.

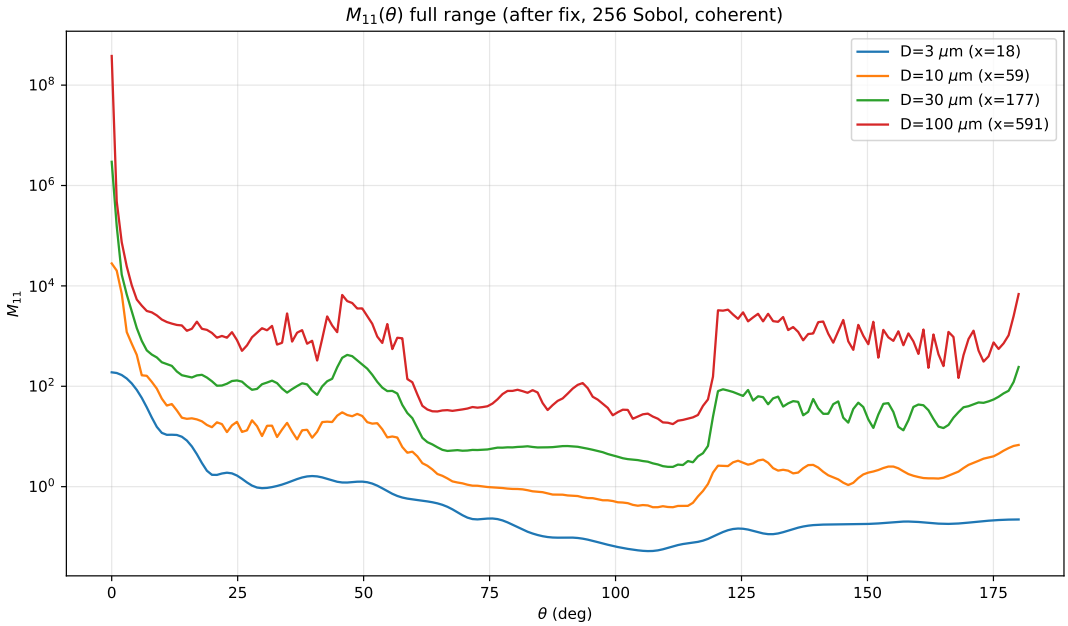

Figure 3: $M_{11}(\theta)$ full range after fix for different particle sizes. Forward peak width scales as $\sim \lambda/D$.

2.4 When Does It Help?

Table 3: Speedup from `-sizefile` vs N separate runs

Scenario	Separate	`-sizefile`	Speedup
N=10 sizes, Phase 1 = 5%	$10 \times T$	$1.5 \times T$	$\sim 7\times$
N=10 sizes, Phase 1 = 50%	$10 \times T$	$5.5 \times T$	$\sim 2\times$
N=2 sizes	$2 \times T$	$1.1 \times T$	$\sim 1.8\times$

Rule of thumb: `-sizefile` helps when Phase 1 (tracing) is expensive relative to Phase 2 (diffraction). This happens for:

- Large n (many reflections): Phase 1 grows, Phase 2 stays constant per size.
- Complex particles (concave, aggregates): tracing is expensive.
- Many sizes ($N \geq 5$): amortization is significant.

It does *not* help when Phase 2 dominates (small n, simple particles)—the overhead of `ComputeFromCache` is similar to `HandleBeamsToLocal` per size.

3 Phi Averaging

3.1 Current Behavior

MBS-raw computes Mueller matrices on a 2D grid (θ, ϕ) where ϕ ranges from 0 to 2π with N_ϕ bins (specified by `-grid`). The output file `M.dat` contains **phi-averaged** $M_{11}(\theta)$: the code sums Mueller over all ϕ bins and divides by $N_\phi + 1$.

3.2 Is There an Auto Mode for Phi?

No. N_ϕ is fixed by the `-grid` argument. There is no `-auto_phi` or adaptive phi mode.

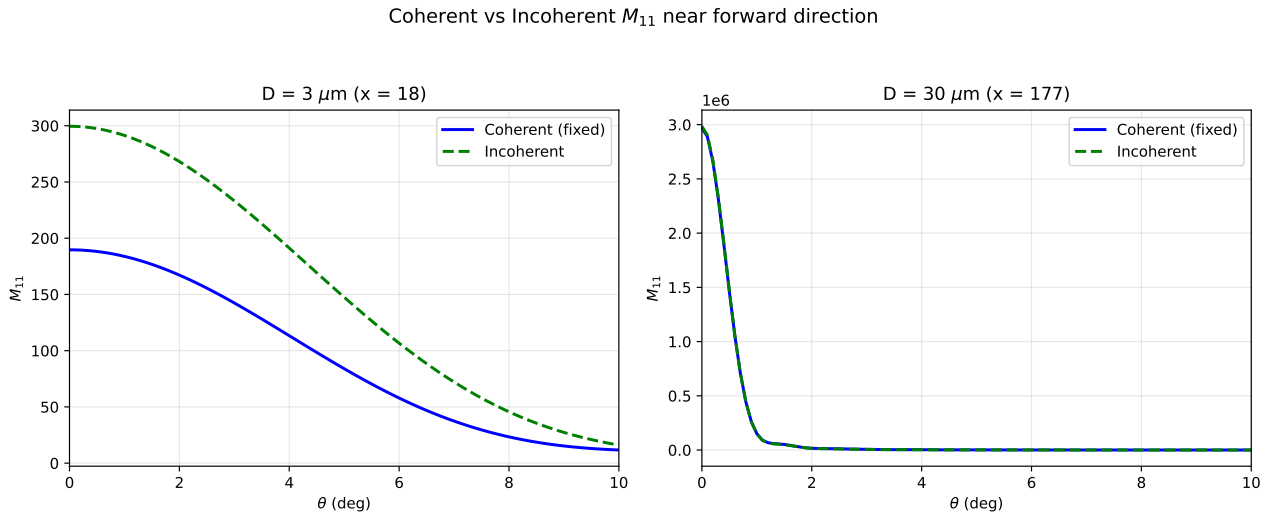

Figure 4: Coherent (fixed) vs incoherent M_{11} near forward direction. For small particles (D=3 μm), both modes agree closely. For larger particles (D=30 μm), coherent forward peak is much sharper.

Table 4: N_ϕ requirements (from validation in previous sessions)

N_ϕ	Error vs N_ϕ=48	Notes
6	50–200% at side/back angles	**Insufficient!** Old default.
12	∼20%	Still poor for polarization
24	$< 5\%$	Acceptable for M_{11}
48	reference	**Recommended minimum**
96	$< 0.1\%$ vs 48	For high-precision work

3.3 How Many Phi Bins Are Needed?

For orientation-averaged calculations, $N_\phi = 48$ is the recommended minimum. $N_\phi = 6$ (the original default) causes gross errors.

3.4 Could Phi Be Optimized?

In principle, fewer ϕ bins could suffice for M_{11} (which varies slowly with ϕ after orientation averaging) while more are needed for M_{33}, M_{34} (fast ϕ oscillations). An adaptive scheme would:

1. Start with $N_\phi = 12$
2. Check M_{11} convergence vs $N_\phi = 24$
3. Double until converged

This is not implemented but would be straightforward. The main barrier is that ϕ bins affect the 2D grid size (memory and computation scale as $N_\phi \times N_\theta$), so reducing N_ϕ gives a proportional speedup.

For most applications, $N_\phi = 48$ is a good fixed default.